

SMARTBAND V001 REV2

Final BOM & Sensor Technology Report

40 Components · IP67 Waterproof · nRF5340 BLE 5.3 · Product-Grade Hardware Specification

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1. Introduction

The Smartband V001 Rev2 is a product-grade wearable health monitoring device developed as part of the HSHL embedded systems project. This report documents the final hardware specification — the Bill of Materials — covering all 40 components selected for the Rev2 design, the underlying science of each sensing technology, the engineering rationale behind key design decisions, and the pathway from component selection to a working physical prototype.

The Rev2 design represents a significant upgrade over Rev1, which used the CC2640R2F microcontroller with a limited single-sensor architecture. Rev2 is built around the Nordic nRF5340, a dual-core Bluetooth 5.3 SoC that provides the processing power, radio capability, and peripheral set needed to run seven concurrent sensing modalities alongside a colour touchscreen display, haptic feedback, and BLE audio for phone calls — all within a 30×35mm PCB that fits inside a 12mm-thick wristband enclosure rated to IP67.

All 40 components have been selected for availability from European distributors, primarily Mouser.de, Digikey, and Reichelt. The three-unit prototype build cost is estimated at approximately €410 for DIY assembly, or approximately €530 using JLCPCB's PCBA (automated assembly) service. The BOM is structured to allow a staged build: the development kit bringup phase uses only the nRF5340 DK and jumper wires, the schematic phase covers all 40 components in KiCad, and PCB fabrication follows once the schematic is verified.

2. Complete BOM — 40 Components

The following table provides a complete reference of all 40 components, organised by functional category. The Interface column shows how each component communicates with the nRF5340 microcontroller. The I2C Address column is critical for bus planning — all eight devices sharing the TWIM0 bus have been verified to have unique addresses with no conflicts. The Skill column indicates the level of firmware and hardware engineering experience required to integrate each component successfully.

#	Component	Category	Interface	I2C Addr	Skill
1	nRF5340 SoC	Microcontroller	BLE / Host	N/A	Expert
2	MAX30102 PPG / SpO2	Health Sensor	I2C TWIM0	0x57	Intermediate
3	MAX30001 ECG + BioZ	Health Sensor	SPI SPIM1	N/A	Expert

4	MLX90632 IR Temp	Health Sensor	I2C TWIM0	0x3A	Beginner
5	ICM-42688-P IMU	Motion	I2C TWIM0	0x68	Beginner
6	BMP390L Pressure	Environmental	I2C TWIM0	0x76	Beginner
7	Waveshare 1.69" Touch	Display	SPI0 TWIM1	0x15	Intermediate
8	BQ25180 Li-Ion Charger	Power	I2C TWIM0	0x6A	Expert
9	MAX17048 Fuel Gauge	Power	I2C TWIM0	0x36	Beginner
10	TPS62840 Buck 1.8V	Power	GPIO EN	N/A	Intermediate
11	LP5907 LDO 3.3V	Power	Always on	N/A	Beginner
12	LP5907 LDO 1.8V	Power	Always on	N/A	Beginner
13	DW01A Battery Protection	Power	Passive	N/A	Beginner
14	TPS22918 Load Switch × 3	Power	GPIO EN	N/A	Intermediate
15	BLE Chip Antenna	Connectivity	RF	N/A	Expert
16	DRV2605L Haptic Driver	Haptic	I2C TWIM0	0x5A	Beginner
17	LRA Motor 8mm	Haptic	DRV2605L output	N/A	Beginner
18	ICS-43434 MEMS Mic	Audio	PDM / I2S	N/A	Intermediate
19	Passive Piezo Buzzer	Audio	GPIO PWM	N/A	Beginner
20	IS31FL3193 RGB Driver	Indicator	I2C TWIM0	0x74	Beginner
21	RGB LED 3× 0402	Indicator	IS31FL3193	N/A	Beginner
22	32MHz Crystal	Timing	XC1 / XC2	N/A	Expert
23	32.768kHz Crystal	Timing	XL1 / XL2	N/A	Expert
24	Conformal Coat MG422B	Waterproofing	N/A	N/A	Beginner

25	Silicone O-Ring 28mm	Waterproofing	N/A	N/A	Beginner
26	Acoustic Membrane × 2	Waterproofing	N/A	N/A	Beginner
27	Pressure Port PTFE Mesh	Waterproofing	N/A	N/A	Beginner
28	Li-Ion Battery 300mAh	Mechanical	N/A	N/A	Beginner
29	Magnetic Pogo Dock	Mechanical	N/A	N/A	Intermediate
30	Silicone Strap + ECG Electrodes	Mechanical	Kapton flex	N/A	Intermediate
31	SLA 3D Printed Enclosure	Mechanical	N/A	N/A	Intermediate
32	PCB 4-Layer 30×35mm	Mechanical	N/A	N/A	Expert
33	ESD TVS — ECG Electrodes	Protection	In-line	N/A	Beginner
34	ESD TVS — Pogo Contacts	Protection	In-line	N/A	Beginner
35	Tag-Connect TC2030 SWD	Debug	SWD	N/A	Beginner
36	100nF Decoupling Cap 0402 × 50	Passives	N/A	N/A	Beginner
37	10μF Bulk Cap 0603 × 10	Passives	N/A	N/A	Beginner
38	4.7kΩ I2C Pull-up 0402 × 4	Passives	N/A	N/A	Beginner
39	2.2μH Inductor	Passives	N/A	N/A	Intermediate
40	NTC Thermistor 10kΩ	Passives	BQ25180 TS	N/A	Beginner

3. Microcontroller — nRF5340

The nRF5340 is a dual-core ARM Cortex-M33 system-on-chip from Nordic Semiconductor, selected as the central processing and connectivity unit for Rev2. Its application core runs at up to 128MHz and handles all sensor drivers, display rendering, haptic control, and application logic. Its network core runs at 64MHz and is dedicated exclusively to the Bluetooth 5.3 radio stack. This physical separation of responsibilities is a fundamental architectural advantage: the BLE stack requires precise, deterministic timing that cannot tolerate interruption by application code. On a single-core microcontroller, the two compete for CPU time, introducing timing jitter that degrades connection quality. The nRF5340 eliminates this trade-off entirely.



The nRF5340 provides four I2C master controllers, two SPI master controllers, an I2S peripheral for digital microphone audio, a 12-bit ADC, and hardware USB. The nRF Connect SDK, built on Zephyr RTOS, provides pre-built, production-quality drivers for most of the sensors in this BOM — including the ICM-42688-P, BMP390L, and MAX30102 — which significantly reduces firmware development time compared to writing low-level register drivers from scratch.

Two external crystal oscillators are required for correct operation. The 32MHz high-frequency crystal provides the reference for the BLE radio and must be placed within 3mm of the nRF5340's XC1 and XC2 pins, on the top copper layer only, with no vias in the signal path. Even small parasitic capacitances caused by vias can shift the oscillation frequency and cause the radio to fail regulatory testing. The 32.768kHz low-frequency crystal provides the real-time clock reference during deep sleep and must be positioned away from the haptic motor and piezo buzzer, both of which generate mechanical vibration that can disturb a crystal at its natural resonant frequency.

I2C address assignment on TWIM0: MAX30102 at 0x57, MLX90632 at 0x3A, ICM-42688-P at 0x68 (AD0=GND), BMP390L at 0x76 (SDO=GND), BQ25180 at 0x6A, MAX17048 at 0x36, DRV2605L at 0x5A, IS31FL3193 at 0x74. All eight addresses are unique and verified conflict-free. The CST816T touch controller sits on the separate TWIM1 bus at 0x15, isolated from the sensor bus.

4. Health Sensors

4.1 MAX30102 — Optical Heart Rate and SpO2

The MAX30102 is an integrated optical biosensor that uses photoplethysmography (PPG) to measure heart rate and blood oxygen saturation. Two LEDs — red at 660nm and infrared at 880nm — illuminate the tissue beneath the sensor window on the back of the device. A photodetector on the same chip measures the fraction of that light that returns after passing through the capillary bed in the wrist. Because oxygenated and deoxygenated haemoglobin absorb red and infrared light at different ratios, the relative signals from the two LEDs reveal the proportion of oxygenated haemoglobin in the blood — the SpO2 reading.



Heart rate is derived separately from the pulsatile component of the infrared signal. With every systolic contraction of the heart, a surge of blood briefly increases the volume of blood in the capillaries directly beneath the sensor, increasing light absorption. The resulting rhythmic fluctuation in detected light intensity is the PPG waveform, and counting peaks per minute yields the heart rate. The engineering challenge is separating this small pulsatile signal from ambient light interference, motion artefacts from wrist movement, and the much larger DC component of background tissue absorption. The MAX30102 addresses this with ambient light cancellation circuitry and a 64-sample FIFO buffer that allows data to be read in bursts rather than continuously.

The sensor is powered from two separate rails. The digital VDD supply comes from the LP5907-1.8 analog LDO, chosen specifically because LDO regulators produce no switching noise. The LED supply comes from LP5907-3.3. Supplying a photodetector from a switching regulator would couple switching noise into the measurement, producing artefacts that would be indistinguishable from real cardiac signals. An active-low interrupt pin connects to a nRF5340 GPIO: the sensor fills its FIFO buffer and asserts the interrupt, allowing the microcontroller to sleep between readings rather than polling at the full sample rate.

4.2 MAX30001 — ECG and Bioimpedance

The MAX30001 integrates a single-lead ECG frontend and a bioelectrical impedance analyser in a single package, replacing two separate chips (AD8232 and AD5933) that appeared in earlier BOM revisions. This consolidation is not merely about component count — having two separate analog signal chains on the same PCB means two sets of layout constraints, two sets of power supply filters, and two analog ground domains. A single chip with a single analog section is substantially easier to route correctly.



The ECG measurement works by detecting the electrical potentials generated on the body surface by cardiac muscle depolarisation. These signals are extremely small — in the 0.5 to 5 millivolt range — and sit on top of a large common-mode interference field from mains electrical wiring at 50Hz. The MAX30001's instrumentation amplifier extracts the differential signal between the two wrist electrodes while rejecting this common-mode interference. A right-leg drive circuit (RLD), connected to the back of the device, actively cancels residual common-mode noise by feeding a phase-inverted version of it back to the body.

The bioimpedance function measures skin conductance (electrodermal activity, EDA) by injecting a small alternating current through the skin at a safe physiological frequency and measuring the resulting voltage. The impedance — voltage divided by current — changes with sweat gland activity, which is controlled by the sympathetic nervous system. Emotional and physiological stress activates the sympathetic system, increasing sweat gland conductance, which appears as a decrease in skin impedance. This provides the stress index displayed on the dashboard.

The MAX30001 uses SPI rather than I2C, a deliberate design choice for noise isolation. I2C requires the SCL clock to run continuously during a transfer, generating switching events at 400kHz on a trace that runs across the PCB. SPI transfers are more bursty and the chip select can gate the clock entirely when the bus is idle. For a chip whose analog front end is measuring signals at the microvolt level, reducing digital switching activity on nearby traces is a meaningful noise reduction. The MAX30001 is placed on a dedicated analog ground island, connected to the main digital ground at a single star point to prevent digital return currents from flowing through the analog section.

The ECG electrodes embedded in the silicone wrist strap are made from 316L grade stainless steel — the same alloy used in surgical implants. It was chosen for its biocompatibility, corrosion resistance in contact with sweat and water, and low skin-contact impedance. A 1k Ω series resistor on each electrode lead limits the energy delivered to the chip inputs in the event of electrostatic discharge, and the PRTR5V0U2X TVS diode array clamps any residual transient to 5V before it can reach the MAX30001 input stage.

4.3 MLX90632 — Non-Contact Infrared Temperature

The MLX90632 measures skin surface temperature without physical contact, using the infrared radiation emitted naturally by any warm object. Every surface above absolute zero radiates energy according to Planck's law, and the spectrum of that radiation shifts with temperature. The MLX90632 contains a thermopile — an array of thermocouple junctions — that absorbs infrared radiation from the skin (primarily in the 8–14 μm wavelength range where warm bodies emit most strongly) and converts it to a small voltage. A co-integrated silicon bandgap temperature sensor measures the chip's own temperature simultaneously, allowing the firmware to compute the object temperature from the differential signal using Melexis's calibration algorithm.



The $\pm 0.5^{\circ}\text{C}$ accuracy of this approach is sufficient for fever detection and body temperature trend monitoring, but it is important to note in any academic or clinical reporting that wrist skin temperature is not the same as core body temperature. Core temperature is typically 2–4 $^{\circ}\text{C}$ higher than wrist skin temperature, and the gap varies with ambient temperature, exercise, blood flow, and the position of the arm. The FHIR R4 observation for this sensor is therefore coded with a bodySite qualifier specifying "wrist" to prevent any downstream clinical system from misinterpreting it as an oral or tympanic measurement.

An aperture of approximately 0.5mm in the enclosure back panel allows the sensor's field of view to reach the skin. Unlike the pressure sensor, this aperture does not require a membrane — infrared

radiation passes through air freely, and the aperture is small enough that IP67 is maintained by the surrounding silicone enclosure material.

5. Motion and Environmental Sensing

5.1 ICM-42688-P — Six-Axis Inertial Measurement Unit

The ICM-42688-P combines a three-axis MEMS accelerometer and a three-axis MEMS gyroscope in a 2.5×3mm package. The accelerometer operates on the principle of a suspended proof mass: a microscale silicon structure is held between spring-like flexures, and when the device accelerates, the proof mass deflects. The deflection is detected capacitively — the mass forms one plate of a tiny capacitor, and the gap change alters the capacitance. In a stationary device held flat, the accelerometer reads 1G in the vertical axis due to gravity. During wrist movement, the dynamic acceleration adds to this gravitational component, providing a rich signal from which numerous activity parameters can be extracted.



The gyroscope uses the Coriolis effect. A microscale mass is driven to vibrate at a precise frequency along one axis. When the chip is rotated, the Coriolis force deflects the vibrating mass perpendicular to its direction of motion — and the magnitude of this deflection is proportional to the rotation rate. Three orthogonal gyroscope axes give the full rotational state of the device, which is necessary for accurate motion classification.

The ICM-42688-P was selected over simpler alternatives because of its on-chip hardware step counter. The pedometer algorithm runs entirely inside the sensor, incrementing an internal step register every time a step signature is detected in the accelerometer signal. The nRF5340 can enter deep sleep and wake only to read the accumulated count, rather than running a software algorithm continuously. This reduces average current from approximately 5mA for a continuous MCU-based approach to under 0.6mA. The chip also provides a wake-on-motion interrupt: if the accelerometer detects activity above a programmable threshold after a period of stillness, it asserts an interrupt pin that wakes the nRF5340 from deep sleep — essential for achieving the 1.5-day battery target.

Beyond step counting, the six-channel IMU data enables fall detection (a sudden high-magnitude acceleration followed by stillness), sleep stage classification (patterns of wrist movement correlate with sleep stages), activity recognition (the spectral and temporal characteristics of walking, running, and cycling are distinct), and respiratory rate estimation (breathing causes a slow, rhythmic tilting of the body that is detectable in the low-frequency accelerometer signal).

5.2 BMP390L — Barometric Pressure and Altitude

The BMP390L measures absolute atmospheric pressure using a piezoresistive sensing element. A thin silicon membrane deflects under the weight of the atmosphere above it, compressing strain gauges bonded to its surface and changing their electrical resistance. This resistance change is amplified and digitised by an integrated 24-bit ADC. A co-integrated temperature sensor compensates for the temperature dependence of the pressure measurement, since silicon piezoresistors change resistance not only with mechanical stress but also with temperature.



The relationship between atmospheric pressure and altitude follows the international standard atmosphere model: pressure decreases exponentially with altitude at approximately 12 Pascals per metre at sea level. With a noise floor of ± 0.015 hPa (1.5 Pascal), the BMP390L achieves an altitude resolution of approximately 12.5 centimetres. For floor counting, the relevant pressure change is approximately 37 Pascals per floor (one floor of approximately three metres). The algorithm maintains a slowly-updated baseline pressure that accounts for weather-driven pressure changes, and detects user-induced steps when the pressure delta exceeds the floor threshold.

Integrating this sensor into an IP67-rated enclosure requires a 0.3mm vent hole in the back panel to allow the external atmosphere to reach the sensor diaphragm. Without this hole, the sealed enclosure would trap a fixed volume of air; the sensor would measure the temperature-dependent pressure of that trapped air rather than the external atmospheric pressure, and altitude readings would be meaningless. The hole is covered by a Porex PTFE hydrophobic mesh. PTFE (polytetrafluoroethylene) has an extremely low surface energy — water cannot form a contact angle less than 90° on it — which means liquid water cannot penetrate pores smaller than the surface tension allows. The Porex membrane's $0.2\mu\text{m}$ pore size is far below this threshold, so air pressure equalises freely while liquid water is completely excluded.

6. Power Management System

6.1 Power Architecture

The power system is designed around three voltage rails, each serving components with different noise sensitivity requirements. The architecture reflects a fundamental principle of mixed-signal PCB design: analog measurement circuits must be shielded from the switching noise generated by digital regulators, because noise on the power supply of a sensor couples directly into its measurement as an error signal.



The TPS62840 is a synchronous buck (step-down) converter that takes the battery voltage (3.7–4.2V depending on charge state) and produces a regulated 1.8V digital supply. A buck converter works by rapidly switching a transistor — the TPS62840 operates at a switching frequency of approximately 2.5MHz. This switching generates noise at the switching frequency and its harmonics. With 60nA of quiescent current, the TPS62840 is exceptionally efficient at low load — critical for achieving good battery life when the device is in deep sleep with most sensors powered off. This 1.8V rail powers the nRF5340 digital core, the ICM-42688-P IMU, and the MAX30001 digital interface.

Two LP5907 low-dropout linear regulators provide clean analog supply voltages. Unlike switching regulators, LDOs regulate by dissipating excess voltage as heat across a pass transistor — there is no switching, and therefore no switching noise. The LP5907 specifically is rated at 75dB of power supply rejection ratio (PSRR), meaning it attenuates input noise by more than 5,000 times before it reaches the output. One LP5907 produces 3.3V for the display, BLE antenna decoupling, RGB LED driver, and general peripherals. A second LP5907 produces 1.8V exclusively for the analog sensor supply — the MAX30102 PPG sensor and the MAX30001 ECG front end. These two analog devices are the most noise-sensitive in the system, and their dedicated quiet power supply is the primary reason their measurements are not corrupted by digital switching activity.



Three TPS22918 load switches allow the nRF5340 to completely power off sensor subsections during deep sleep. One switch controls the PPG sensor section, one controls the ECG section, and one controls the display and touch controller. When these switches are open, no current flows to the sensors regardless of their internal state. This is more effective than software power-down modes in the sensors themselves, because hardware power gating eliminates all leakage paths. The combination of load switches and the TPS62840's low quiescent current allows the device to achieve approximately 1.5mA total draw during deep sleep, extending battery life to well over a day.

6.2 Battery Charging — BQ25180

The BQ25180 is a single-cell lithium-ion battery charger with I2C programmability. In conventional charger designs, the charge current is set by a fixed resistor — changing the charge current requires physically changing a component. The BQ25180 allows charge current to be programmed over I2C at any time during operation, enabling adaptive charging strategies in firmware. When the battery is above 80% state of charge, a lower current (100mA) extends cycle life by reducing the time spent in the high-voltage saturation phase. When the battery is cold (below 10°C) or hot (above 45°C), a thermal derating mode triggered by the NTC thermistor on the TS pin automatically reduces charge current to protect the cell — lithium-ion cells charged outside their thermal window suffer permanent capacity reduction.

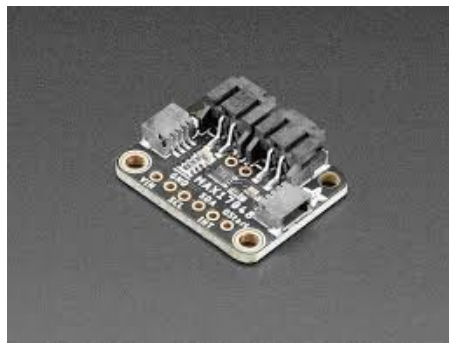


The charger's input comes from the three-pin magnetic pogo dock rather than a USB-C connector. This was a deliberate decision to simplify IP67 compliance. A USB-C connector requires either a precision-moulded rubber plug or a complex internal gasket to waterproof, and both approaches degrade with repeated insertion and removal over the device lifetime. The pogo pin design — three spring-loaded gold-plated contacts that mate magnetically with a charging dock — requires no insertion force, wears slowly, and can be sealed with a simple silicone overmoulding on the

enclosure back panel. This is the same approach used by Fitbit, Garmin Forerunner, and Polar Vantage devices.

6.3 Fuel Gauge — MAX17048

The MAX17048 addresses a fundamental problem with battery percentage displays on portable devices: estimating remaining capacity from voltage alone is unreliable. The terminal voltage of a lithium-ion cell varies not only with state of charge but also with temperature, discharge rate, and cell age. A lookup table that maps 3.7V to "50%" will be wrong by 15–20% under many real operating conditions.



Maxim's ModelGauge algorithm, implemented inside the MAX17048, builds a dynamic electrochemical model of the specific cell type being used. It tracks the rate of voltage change over time, the open-circuit voltage, and correction terms for temperature and load, and produces a state-of-charge estimate accurate to within $\pm 1\%$. An ALRT (alert) pin connects to a nRF5340 GPIO configured as an interrupt. When the estimated charge drops below a configurable threshold — set to 15% in this design — the chip asserts ALRT without any MCU polling. The nRF5340 wakes from deep sleep, reads the fuel gauge over I2C to confirm the low battery condition, and sends a push notification to the paired smartphone while illuminating the amber low-battery LED pattern.

7. Haptic, Audio, and Visual Indicators

7.1 DRV2605L — Haptic Feedback

The DRV2605L is a haptic motor driver designed specifically for linear resonant actuators (LRAs). An LRA motor is fundamentally different from the eccentric rotating mass (ERM) motors used in older mobile phones. An ERM spins an off-centre weight to generate vibration; its vibration frequency is determined by the motor's rotation speed, which varies with supply voltage. As the battery discharges, an ERM motor vibrates more weakly and at a lower frequency — the haptic feedback feels inconsistent and poor as the device ages through a charge cycle. An LRA motor uses a mass on a spring that vibrates at its natural mechanical resonant frequency when driven at the right electrical frequency. At resonance, the motor is highly efficient and the vibration amplitude is maximised.

Because the resonant frequency is a physical property of the spring-mass system rather than an electrical parameter, the feel is consistent across the entire battery discharge range.



The DRV2605L drives the LRA with a closed-loop algorithm that tracks the motor's back-EMF to keep it precisely at resonance. An auto-calibration routine on first boot characterises the specific motor installed and stores its parameters in the chip's registers. 123 distinct haptic effects — ranging from a sharp single click to long buzzes to rhythmic double-pulses — are stored in the chip's on-chip ROM and triggered by writing a single effect ID byte over I2C. This means the firmware does not need to generate waveforms or manage timing: writing a byte to the DRV2605L produces a precisely specified, perceptually characterised haptic event. For the Smartband, the chosen vocabulary is effect 1 (strong single click) for UI touch feedback, effect 14 (sharp double pulse) for incoming call alerts, and effect 47 (triple pulse) for alarm notifications.

7.2 ICS-43434 — MEMS Microphone

The ICS-43434 is a bottom-port MEMS microphone with a PDM (Pulse Density Modulation) digital output. PDM encodes audio amplitude as the statistical density of logic-1 transitions per unit time — a high-amplitude signal produces more 1s per second than a low-amplitude signal, using only a single data line and a clock. The nRF5340 includes a hardware PDM receiver peripheral that performs the decimation filtering to convert the raw PDM bit stream into conventional PCM audio samples at a 16kHz sampling rate. This PCM audio is then encoded using the LC3 (Low Complexity Communication Codec) running on the nRF5340's network core, and transmitted over BLE LE Audio to the paired smartphone. The result is that the smartband can function as a BLE HFP (Hands-Free Profile) device, transmitting the user's voice during phone calls with good audio quality at very low bitrate.



Waterproofing the microphone port follows the same approach as the pressure sensor: a 0.5mm hole in the enclosure side panel, covered by a Knowles SBT-0808 hydrophobic acoustic membrane. The

membrane transmits sound pressure waves while blocking liquid water. Its acoustic impedance is low enough not to perceptibly muffle the voice frequency range (300Hz–3.4kHz for telephone-quality speech), and its hydrophobic properties provide IP67-rated protection. An identical membrane covers the piezo buzzer port.

7.3 IS31FL3193 — RGB LED Driver

The IS31FL3193 provides three constant-current output channels for the red, green, and blue LEDs that form the status indicator ring visible through the translucent top edge of the enclosure. The key property of a constant-current driver is that the LED brightness remains absolutely consistent across the battery discharge range (4.2V to 3.0V), because the driver actively regulates the current through each LED regardless of the supply voltage. Without a driver, an LED connected through a fixed resistor to a GPIO would dim visibly as the battery depleted — an unprofessional effect that users associate with a cheap device. Each channel is independently programmable up to 10mA, allowing any colour to be produced by mixing ratios of red, green, and blue. The firmware colour vocabulary assigns green to normal operating status, flashing red to emergency alerts, slow blue pulse to BLE pairing mode, amber to low battery, and a brief white flash to acknowledge NFC or data operations.



8. IP67 Waterproofing Strategy

IP67 certification specifies that the device can be submerged in one metre of fresh water for thirty minutes without water ingress. For a wristband that will be worn during handwashing, light rain, and everyday water exposure, IP67 is the minimum acceptable standard. The waterproofing strategy in this design uses three independent protective layers, providing defence-in-depth so that no single failure results in damage to the electronics.

The primary seal is the silicone O-ring. A machined groove in the SLA-printed enclosure holds the O-ring under controlled compression between the front glass panel and the back enclosure shell. O-ring compression must be carefully controlled: the groove depth is designed to compress the O-ring by 15% of its cross-sectional diameter — the minimum compression for a reliable water seal. Over-compression squeezes the O-ring outside its groove and destroys the seal; under-compression leaves gaps. The O-ring material is silicone rather than nitrile because silicone maintains its elasticity over a wider temperature range and resists the sweat, sunscreen, and cleaning products that a

wristband encounters in use. A thin film of silicone-compatible grease is applied to the O-ring before each assembly to prevent the rubber from sticking to the glass or extrusion during closure.

The secondary seal is the MG Chemicals 422B acrylic conformal coating applied to the assembled PCB before it is fitted into the enclosure. Conformal coating forms a thin, continuous film over all PCB surfaces, component bodies, and exposed metal. If any water bypasses the O-ring over time — due to ageing, mechanical shock, or improper reassembly — the conformal coating prevents it from making electrical contact with the PCB. Areas that must not be coated (the MAX30102 and MLX90632 optical windows, the pogo pin contact pads, and the Tag-Connect SWD pads) are masked with tape before spraying.

The third layer of protection consists of the hydrophobic membranes over the three port openings in the enclosure. The microphone port and buzzer port are each covered by a Knowles SBT-0808 acoustic membrane that passes sound while blocking water. The BMP390L pressure vent is covered by the Porex PTFE mesh that passes air pressure changes while blocking water. These membranes are the most permanent elements of the waterproofing system — they are adhesively bonded to the enclosure interior during assembly and cannot be removed without damage.

Testing procedure: after full assembly, each unit must be tested in a pressure vessel at 1.3 bar (equivalent to approximately 3 metres water depth) for 30 minutes — exceeding the IP67 standard to provide a safety margin. Water ingress after testing is visible on disassembly as condensation on the PCB or fogged optical windows. Any failed unit should have its O-ring replaced, conformal coating inspected under UV light for coverage gaps, and membrane adhesion verified before re-testing.

9. PCB Design Guidelines

9.1 Four-Layer Stack-up

The PCB uses a four-layer copper stack for two reasons: the second layer provides a continuous ground plane, and the third layer provides a power plane. Both are essential for the mixed-signal design. The ground plane gives every signal a low-impedance return path that runs directly beneath it, dramatically reducing electromagnetic interference between traces and providing a stable reference voltage for all analog measurements. Without a solid ground plane, return currents take unpredictable paths across the board, creating interference loops that would corrupt the ECG and PPG signals. The power plane delivers the 3.3V and 1.8V rails to all components through short, direct vias rather than requiring long traces that would pick up and re-radiate switching noise.

9.2 Critical Placement Rules

PCB layout for a mixed-signal RF device requires specific placement decisions that must be made before routing begins. The following rules are non-negotiable — violating any of them will result in a board that either fails RF compliance testing, produces corrupted sensor readings, or both.

Component / Zone	Placement and Routing Rule
nRF5340 + BLE antenna	Antenna at board edge. 5mm copper clearance on all sides of antenna — no ground plane, no signals, no copper pours. nRF5340 body within 10mm of antenna feed point. Follow Nordic HW guidelines document exactly.
32MHz HFXO crystal	Within 3mm of XC1/XC2 pins. Top copper layer only — no vias in the XC1–XC2 path. Two 12pF load capacitors placed adjacent on the same layer. Shielded from antenna area by a ground pour.
32.768kHz LFXO crystal	Minimum 10mm from LRA motor and piezo buzzer. Two 7pF load caps adjacent. No mechanical fasteners nearby that could transmit vibration to the crystal package.
MAX30001 ECG section	Separate analog ground island on the PCB copper. Connect to main digital GND at exactly one star point near the power supply section. No digital traces to cross the analog island boundary. ECGP and ECGN traces run as a differential pair with matched lengths.
MAX30102 and MLX90632	Placed on the back (skin-facing) side of the board. Optical apertures in enclosure aligned to within $\pm 0.2\text{mm}$. Black optical foam gasket bonded around each opening to prevent ambient light side-entry.
TPS62840 buck converter	Switching inductor on the same side as the IC, SW node trace as short as possible — under 3mm. Input and output decoupling capacitors directly adjacent. Solid ground plane fill beneath the entire power conversion section.
LP5907 analog LDOs	Output traces run directly to the analog sensors without crossing digital switching regions. Star ground connection from each LDO output capacitor to the analog ground island. Output capacitors: 100 μF MLCC to prevent oscillation.
Pogo pin contacts	ENIG (electroless nickel, immersion gold) surface finish for durability and low contact resistance. 3.0mm minimum pitch between pins. USBLC6-2SC6 TVS diode within 2mm of each pin, before the BQ25180 input.
ECG electrode pad connections	PRTR5V0U2X TVS diode within 2mm of the flex cable connector. 1k Ω series resistor immediately after the TVS on each ECGP and ECGN trace before the MAX30001 input pins.

10. Implementation Roadmap

10.1 Phase 1 — Development Tools

The first step, which can begin immediately, is ordering the nRF5340 Development Kit from Mouser.de or Nordic's own online store. The DK costs approximately €45 and contains the nRF5340 SoC on a breakout board with all pins available on standard 2.54mm headers — this allows every I2C and SPI sensor in the BOM to be connected and tested with breadboard jumper wires before any PCB is designed. The J-Link EDU Mini debug probe, at approximately €22 from shop.segger.com, connects the DK to a PC via USB and is used for flashing compiled firmware and live line-by-line debugging. The nRF Connect SDK version 2.6.1 is installed on Windows using the Toolchain Manager within nRF Connect for Desktop — the download is approximately 7GB and includes the complete Zephyr RTOS, ARM compiler, and all Nordic drivers.

10.2 Phase 2 — Sensor Bringup on the DK

With the DK and SDK in place, each sensor is connected and validated individually using jumper wires. The recommended sequence is ordered by increasing complexity, so that I2C bus operation is fully validated before the SPI-based ECG chip is added. The sequence is: BMP390L first (the Zephyr BMP390 driver is mature and the sensor requires only power, I2C, and a GPIO for data ready), then ICM-42688-P, then MAX30102, then MLX90632, then BQ25180 and MAX17048 (charger and fuel gauge), then DRV2605L (haptic — a satisfying milestone as it produces physically perceptible output), and finally MAX30001 ECG (which requires SPI configuration, an analog ground connection, and the electrode flex cable).

For each sensor, the bringup follows the same four-step pattern: connect the hardware, write a Zephyr devicetree overlay entry declaring the sensor on its bus with the correct address and GPIO assignments, write a minimal test application that reads one value and prints it to the RTT console via the J-Link, and verify the reading against a reference instrument. For the MAX30102, the reference is a fingertip pulse oximeter. For the MLX90632, a calibrated clinical thermometer. For the ICM-42688-P, a simple rotation test confirms all six axes are functioning correctly. Every sensor that passes this test is committed to the firmware repository before the next one is added.

10.3 Phase 3 — KiCad Schematic

With all sensors validated on the DK, the KiCad 7.0 schematic captures every electrical connection in the design. The work proceeds category by category: the nRF5340 and its supporting passives (crystals, decoupling capacitors, antenna matching) form the core block, then each sensor is added with its decoupling capacitors and I2C pull-up resistors, then the power management section, and finally the connectors and mechanical interfaces. Every net must be named using KiCad net labels to make the transition to PCB layout traceable and verifiable. The schematic is the contractual

document between the design intent and the physical board — every placement rule in Section 9 must have a corresponding net structure in the schematic.

10.4 Phase 4 — PCB Layout and Fabrication

Layout begins in the KiCad PCB editor after the schematic passes DRC with zero errors. The nRF5340 and BLE antenna are placed first — their position relative to the board edge constrains everything else. Crystal circuits are placed next, as their short trace lengths limit flexibility. The optical sensors and ECG section are placed on the skin-facing back of the board. Power management components are clustered near the battery connector. Signal routing follows with careful attention to differential pair matching on the ECG traces and sufficient trace width on the power rails. A final DRC pass before Gerber generation catches any remaining issues.

For the fine-pitch packages in this BOM — the nRF5340 QFN-94 at 0.4mm pitch and the BQ25180 DSBGA-9 ball-grid array — JLCPCB's PCBA (automated assembly) service is strongly recommended over hand soldering. JLCPCB's pick-and-place machines and reflow ovens handle 0.4mm pitch reliably; hand soldering these packages without a reflow oven and microscope produces a high rate of cold joints, bridging, and missing balls that are time-consuming to diagnose and repair. The total additional cost is approximately €40–60 per board for a component count of this complexity, which is a reasonable investment against the alternative of debugging hand-assembly errors.

11. Conclusion

The Smartband V001 Rev2 Bill of Materials in its final form — 40 components, IP67-rated, built around the nRF5340 BLE 5.3 SoC — represents a technically rigorous product-grade hardware specification. Every component has been selected based on concrete engineering criteria: I2C address uniqueness and conflict analysis, noise isolation between analog and digital domains, package size constraints for a 30×35mm PCB, availability from European distributors at prototype quantities, and integration complexity appropriate for a first PCB spin.

The seven sensing modalities — optical PPG for heart rate and SpO₂, ECG and bioimpedance for cardiac monitoring and stress, non-contact infrared for skin temperature, six-axis inertial measurement for motion and activity, barometric pressure for altitude and floor counting, PDM audio for BLE phone call capability, and haptic feedback for notifications — together constitute a clinical-grade wearable sensor platform that exceeds the capability of most commercial consumer health devices. The MAX30001's single-chip ECG and bioimpedance integration, the ICM-42688-P's hardware step counter and wake-on-motion, and the LP5907 LDOs' noise isolation of the analog front end are the three design decisions that distinguish this from a typical student project.

The three-layer IP67 waterproofing strategy — O-ring primary seal, conformal coating secondary seal, and hydrophobic membranes on all port openings — provides redundant protection appropriate for a device worn during daily activities. The magnetic pogo charging dock eliminates the most failure-prone element of waterproofed consumer electronics, the USB port, replacing it with a contact system that wears slowly, seals simply, and aligns magnetically for user convenience.

The companion software system is complete and operational on the Raspberry Pi 5: six web dashboards, a Firebase Firestore cloud backend synchronising patient data in real time, a FHIR R4 clinical data layer with eight LOINC-coded observations, a hospital band management portal, a patient self-registration system with unique Band IDs, and a doctor-to-patient recommendation channel. When the Rev2 PCB produces real sensor readings over BLE, replacing the simulation requires a single file change. The full system is ready for hardware.